

BASTIN PRASAD

Software Engineer | .NET | LIS/RIS | Healthcare Integration

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PROFESSIONAL SUMMARY

Software Engineer with nearly two years of professional experience developing and supporting healthcare software, specializing in Laboratory Information Systems (LIS), Hospital Information Systems (HIS/EMR) and Radiology Information Systems (RIS). Experienced in C#, ASP.NET, SQL Server, healthcare integrations, laboratory analyzer integration, HL7/ASTM communication, database development, production support and troubleshooting across application, database, device and network layers. Currently contributing to the modernization of an existing LIS into a modern .NET architecture.

TECHNICAL SKILLS

Languages: C#, VB.NET, T-SQL, JavaScript, HTML, CSS

.NET & Backend: ASP.NET Web Forms, ASP.NET Core, .NET Worker Services / Windows Services, REST APIs, SignalR, JWT Authentication

Database: Microsoft SQL Server, Stored Procedures, Complex Queries & Joins, Query Optimization, Schema Changes, Database Migration, Data Validation

Healthcare Systems & Integration: LIS, RIS, HIS/EMR, HL7, ASTM, Laboratory Analyzer Integration, Device/Test/Service Mapping, Worklists, Result Processing

Networking & Communication: TCP/IP, Serial/COM Communication, Serial-to-Network Devices (Moxa), Network Troubleshooting (ping, Test-NetConnection, netstat)

Frontend: Angular, HTML, CSS, JavaScript

Tools: Visual Studio, SQL Server Management Studio, Git, Swagger, Insomnia, UiPath

AI Tools: ChatGPT, Claude, GitHub Copilot (AI-assisted development, debugging and documentation)

PROFESSIONAL EXPERIENCE

Associate Software Engineer — QDES Infotech Pvt. Ltd., Kochi, India

Nov 2024 – Present

- Develop, maintain and enhance a Laboratory Information System (LIS) using C#, ASP.NET and SQL Server, supporting lab workflows including worklists, result processing, result logging and report generation.
- Integrate laboratory analyzers such as Sysmex XN-550, Erba XL-200, Cobas e411 and Atellica UAS 1500, handling device configuration, device/test/service mapping, communication setup and result import.
- Implement and troubleshoot HL7- and ASTM-based communication workflows between laboratory instruments and the LIS.
- Work on Radiology Information System (RIS) integration, including radiology worklists, study updates, service/test mapping and RIS database processing.
- Supported HIS/EMR applications, troubleshooting application, database and integration-related issues and coordinating issue resolution across hospital workflows.
- Develop and maintain SQL Server stored procedures and queries for LIS/RIS workflows such as worklist retrieval, service and test mapping, study updates and result logging.
- Perform database migration, schema and procedure updates, and data validation across client healthcare environments.
- Investigate and resolve production issues across application, database, device and network layers, improving application stability.
- Troubleshoot TCP/IP and serial/COM connectivity for analyzers, and investigate virtual COM and serial-to-network communication architectures.
- Optimize SQL queries and backend processes to improve performance and reliability.
- Collaborate with implementation and support teams to diagnose site-specific configuration and integration issues.

ASP.NET Developer Intern — Luminar Technolab, Kochi, India

Jun 2024 – Nov 2024

- Developed web applications using ASP.NET Web Forms, C# and SQL Server.
- Worked on UI development, backend logic, database operations and debugging.

KEY PROJECTS

LIS Platform & Modernization

Legacy LIS → LIS.Next | C#, ASP.NET, SQL Server → .NET 10, ASP.NET Core, Angular 22, SignalR, JWT

- Maintained and enhanced an existing LIS application developed using C#, ASP.NET and SQL Server, supporting laboratory workflows and production deployments.
- Worked on application maintenance, debugging, database changes, stored procedures, device/test mapping, analyzer integration and production issue resolution.
- Contributed to the modernization of the existing LIS into LIS.Next, transitioning toward a modern .NET architecture.
- Designed a modern LIS architecture using .NET 10 Worker Services and ASP.NET Core REST APIs.
- Explored Angular 22 frontend architecture using standalone components and signals, with JWT authentication and SignalR for real-time communication.
- Developed an analyzer simulator and regression-testing approach to validate integration workflows without requiring physical laboratory instruments.

Radiology Integration System (RIS)

C#, SQL Server, RIS, HL7

- Worked on Radiology Information System integration and backend components supporting radiology workflows.
- Developed and modified SQL Server stored procedures for radiology worklists, service details, study updates and test/service mapping.
- Worked on integration between RIS components and hospital databases.
- Troubleshoot data-flow and configuration issues across application and database layers.

EDUCATION

Bachelor of Computer Applications (BCA)

2021 – 2024

Santhigiri College of Computer Sciences, Thodupuzha

Academic projects: Construction Management System (ASP.NET), Auditorium Booking Portal (PHP)

Higher Secondary Education — Holy Cross HSS, Cherpunkal

2021

CERTIFICATIONS

ASP.NET Full Stack (5 months, Grade A+) — NACTET, through Luminar Technolab

Nov 2024

Cloud Computing (8-week course) — IIT Kharagpur

Oct 2022

Introduction to Cybersecurity — Cisco Networking Academy

Responsive Web Design — freeCodeCamp

LANGUAGES

English, Malayalam